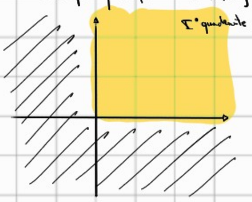
① dominio :

$f(x,y) = \sqrt{x} - \sqrt{y}$ $\quad$ $dom(f) = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0\}$

$g(x,y) = \sqrt{x} - \frac{1}{\sqrt{y}}$

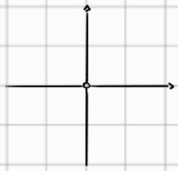


↑ stesso ma con $y \neq 0$

$dom(g) = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0, y > 0\}$

② dominio e curve di livello

$f(x,y) = \frac{1}{x^2+y^2}$ $\quad$ $dom(f) = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \neq 0\}$
$= \mathbb{R}^2 \setminus \{0,0\}$



$\mathcal{L}(f,k) = \{(x,y) \in dom(f) \mid f(x,y) = k\}$
$= \{(x,y) \in \mathbb{R}^2 \mid \frac{1}{x^2+y^2} = k\}$ (*)

Se $k \leq 0$ allora $\mathcal{L}(f,0) = \emptyset$

Se $k > 0$ allora (*) diventa $x^2+y^2 = \frac{1}{k}$

e quindi $\mathcal{L}(f,k) = \partial B(0, \frac{1}{\sqrt{k}})$

Rem

$y = mx+q$ retta

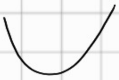
$x^2+y^2 = k$ circonferenza

$y = ax^2+bx+c$ parabola

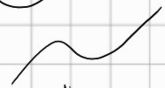
$y = 5x^4$

$x^2+2y^2 = \dots$ elisse iperboloide

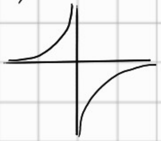
$y = x^{2m}$



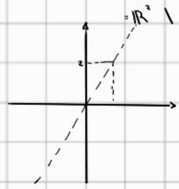
$y = x^{2m+1} + \dots$



$xy = 1$



③ $f(x,y) = \frac{2x-y+1}{2x-y}$ $\quad$ $dom(f) = \{(x,y) \in \mathbb{R}^2 : 2x-y \neq 0\}$
$= \mathbb{R}^2 \setminus \{\text{retta } y = 2x\}$



$\mathcal{L}(f,k) = \{(x,y) \in dom(f) : \frac{2x-y+1}{2x-y} = k\}$

$2x-y+1 = 2kx - ky$

$(k-1)y - 2(k-1)x + 1 = 0$

Se $k \neq 1$ $\quad y = 2x - \frac{1}{k-1}$ | Se $k = 1$ $\quad 1 = 0$ FALSO

$\mathcal{L}(f,k) = \begin{cases} \text{retta di eq.} \\ y = 2x - \frac{1}{k-1} \end{cases}$ | $\mathcal{L}(f,1) = \emptyset$

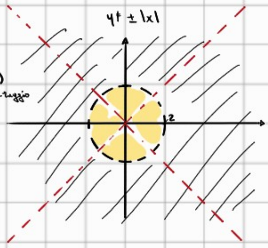
$Im(f) = \mathbb{R} \setminus \{1\}$

④ $f(x,y) = \frac{\ln(4-x^2-y^2)}{|x|-|y|}$

$dom(f) = \{(x,y) \in \mathbb{R}^2 : \overset{x^2+y^2<4}{4-x^2-y^2 > 0}$
$\& \ |x|-|y| \neq 0\}$
↓
$y \neq \pm|x|$

Ricordiamo che

$x^2+y^2 = 4$ $\quad \partial B(0,2)$
↑ indice ↑ raggio



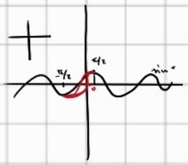
$y = |x|$

$y = -|x|$

⑤ $f(x,y) = \frac{\arcsin(x^2-y)}{xy}$

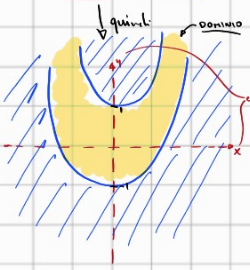
$dom(f) = \{(x,y) \in \mathbb{R}^2 : xy \neq 0\}$

① $xy = 0$? $\quad x = 0$ oppure $y = 0$



$\underline{xy \neq 0}$ $\quad \underline{x \neq 0}$ & $\underline{y \neq 0}$

↓ quindi



$\sin : [-\frac{\pi}{2}; \frac{\pi}{2}] \to [-1,1]$ è invert

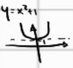
$\arcsin : [-1,1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$ è l'inverso

$-1 \leq x^2-y \leq 1$

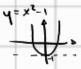
$\begin{cases} x^2-y \leq 1 \\ x^2-y \geq -1 \end{cases}$ $\quad \begin{cases} y \geq x^2-1 \\ y \leq x^2+1 \end{cases}$

→ $y = x^2$ →

→ $y = x^2+1$



→ $y = x^2-1$



$f(x,y) = \sqrt{\sin(x^2+y^2)}$

$dom(f) = \{(x,y) \in \mathbb{R}^2 : \sin(x^2+y^2) \geq 0$

è positivo solo qui

$\sin T > 0$

$0 + 2k\pi < T < \pi + 2k\pi$

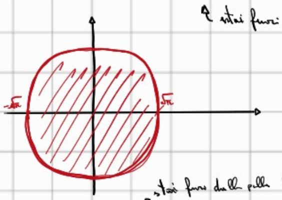
$2k\pi \leq x^2+y^2 \leq \pi + 2k\pi$

per qualche $k \in \mathbb{N}$

con $\underline{k=0}$ $\quad$ verificato pk somma di quadrati ↓ $\cancel{0 \leq} \ x^2+y^2 \leq \pi$

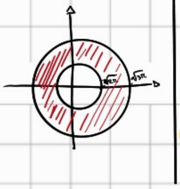
→ $\overline{B(0,\sqrt{\pi})}$

↑ stai fuori dalla palla



stai fuori dalla palla $B(0,\sqrt{2\pi})$

con $\underline{k=1}$ $\quad 2\pi \leq x^2+y^2 \leq 3\pi$



⑥ $f(x,y) = \ln(xy^2+x^2y)$

$dom(f) = \{(x,y) \in \mathbb{R}^2 \mid xy^2+x^2y > 0\}$
↓
$xy(y+x) > 0$

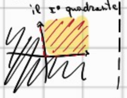
quand'è che un prodotto di 3 cose è positivo?

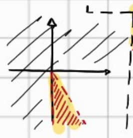
→ tutti positivi
→ 1 positivi e 2 negativi
} tutte le possibilità

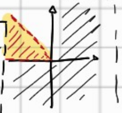
I° caso $\quad$ OPPURE OR

$\begin{cases} x > 0 \\ y > 0 \\ \underline{y+x > 0} \end{cases} \cup \begin{cases} x > 0 \\ y < 0 \\ x+y < 0 \end{cases} \cup \begin{cases} x < 0 \\ y > 0 \\ x+y < 0 \end{cases} \cup \begin{cases} x < 0 \\ y < 0 \\ x+y > 0 \end{cases} \to \emptyset$

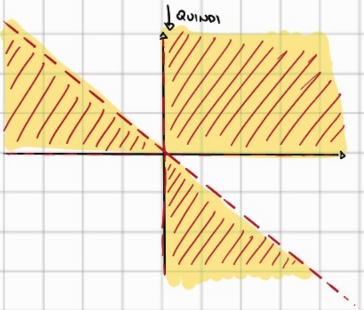
↓ la soluzione è il I° quadrante







↓ QUINDI



Dominio e Curve di livello

$f(x,y) = \frac{x^2}{y^2}$ $\quad$ $dom(f) = \{(x,y) \in \mathbb{R}^2 \mid y \neq 0\}$
$= \mathbb{R}^2 \setminus \{\text{asse } x\}$

$\mathcal{L}(f,k) = \{(x,y) \in dom(f) : \frac{x^2}{y^2} = k$

Se $k < 0$ allora $\mathcal{L}(f,k) = \emptyset$

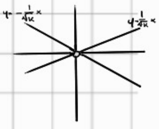
Se $k = 0$ allora $x^2 = 0 \to x = 0$

$\mathcal{L}(f,0) = \{\text{asse } y\} \setminus \{(0,0)\}$

Se $\underline{k > 0}$ allora $x^2 = ky^2$ $\quad y^2 = \frac{1}{k} - x^2$

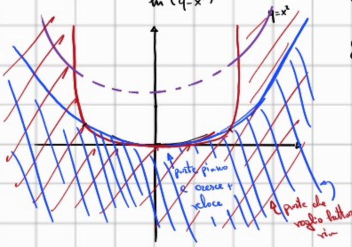
$y = \pm \frac{1}{\sqrt{k}} x$

↑ è una retta



$\mathcal{L}(f,k) = \left(\left\{ \begin{matrix} \text{retta} \\ y = \frac{1}{\sqrt{k}}x \end{matrix} \right\} \cup \left\{ \begin{matrix} \text{retta} \\ y = -\frac{1}{\sqrt{k}}x \end{matrix} \right\} \right) \setminus \{(0,0)\}$

⑨ $f(x,y) = \frac{\sqrt{y-x^4}}{\ln(y-x^2)}$ $\quad$ $dom(f) = \{(x,y) \in \mathbb{R}^2 \mid$
$\underline{y \geq x^4}$ → paraboloide
& $\underline{y > x^2}$ → parabola
& $\underline{y - x^2 + 1}\}$
→ buco di parabola



$f(x,y) = \int_2^{|x|-|y|} \frac{e^t}{(t+1)\sqrt[3]{t}} dt$ $\quad$ $g(t) = \frac{e^t}{(t+1)\sqrt[3]{t}}$ → $dom(g) = (-\infty,-1) \cup (-1,0) \cup (0,+\infty)$

$t \to -1$: $\quad$ $dom\ f = ?$ $\quad$ $G(z) = \int_2^z g(t)\,dt$

$g(t) \approx \frac{e^{-1}}{(t+1)^{(-1)}} \xrightarrow{t \to -1} \infty$ di ordine 1 → quindi l'$\int$ $\underline{\text{DIV.}}$

$t \to 0$ $\quad g(t) \sim \frac{1}{\sqrt[3]{t}} = t^{-1/3} \to \infty$ di ordine $1/3 < 1$ e quindi l'$\int$ $\underline{\text{conv.}}$

$dom(f) = \{(x,y) \in \mathbb{R}^2 \mid |x|-|y| > -1\}$ → $\begin{cases} y < |x|+1 \\ y > -|x|-1 \end{cases}$